



Mock Test #4

Complete Structured Notes | 20 Questions Covered

Biology · History · Polity · Geography · Economics · Physics



① Biology — Cell Biology (Cytology)

Q1. What is the primary function of the Golgi apparatus?

- ☒ A Protein modification, packaging and dispatching ✓
- ☐ B DNA replication
- ☐ C ATP production
- ☐ D Lipid synthesis

✓ Answer: A — Golgi Apparatus = "Post Office of the Cell"

→ Key Discoveries — Cell History

DISCOVERY TIMELINE

1665 → Robert Hooke → Dead cell (cork)

1674 → Antony van Leeuwenhoek → Living cells (pond water)

CELL THEORY

Schleiden + Schwann → Every living being made of cells

Modified by Virchow → New cell forms from pre-existing cell

→ Cell Size Reference

Mycoplasma

Smallest cell overall

Ostrich egg

Largest cell overall

Sperm → Ovum

Smallest → Largest (humans)

Neuron

Longest cell

→ Cell Membrane vs Cell Wall

Feature	Cell Membrane	Cell Wall
Nature	Living	Dead
Permeability	Semi-permeable / Selectively permeable	Completely permeable
Composition	50% Protein + 50% Lipid	Plants = Cellulose Bacteria = Peptidoglycan Fungi = Chitin

→ Cell Organelles – Complete Guide



Golgi Apparatus

Protein modification + packaging + dispatching

"Post Office of the Cell"

Lysosomes formed here (enzymes made in RER)



Endoplasmic Reticulum

RER → has ribosomes → synthesises proteins + hydrolytic enzymes

SER → synthesises fats/lipids + detoxification



Ribosomes

Discovered by **George E. Palade**

No membrane | Present in BOTH prokaryotes + eukaryotes

Common features of both: Ribosome + Cell membrane + Cytoplasm



Mitochondria

ATP production (Adenosine Triphosphate = energy currency)

Has **own DNA and ribosomes**

2 membranes: Outer (porous) + Inner (cristae)

Krebs cycle → **matrix** | 36 ATP → **inner membrane**



Lysosomes

Contain hydrolytic/digestive enzymes

Burst → digests entire cell → **"Suicidal Bags"**

Formed in Golgi; enzymes made in RER



Centrioles

Help in cell division

MTOC = Microtubule Organising Center

Centrosome = MTOC | Arrangement: 9+0 or 9+2

→ Nucleus – Complete Facts

- Contains **nucleolus** inside
- Chromatin (thread-like) → condenses during division → **chromosomes**
- Humans: **23 pairs** of chromosomes | 22 pairs = Autosomal | 1 pair = Sex (XX = Female, XY = Male)
- Chromosomes made of DNA + Protein (**Histones**)
- Functional segment of DNA = **Gene**

FORMULA

DNA Replication Enzymes (in order):

1. Helicase (unwinds) → 2. Primase (initiates) → 3. Polymerase (builds) → 4. Ligase (joins/glues)

DNA Double Helix → Watson & Crick (1953) | A-T | G-C pairs

→ Plastids – Plant Cells Only

Plastid Type	Function
Chloroplast	Makes food (photosynthesis)
Chromoplast	Provides colour (anthocyanins, carotenoids)
Leucoplast	Stores food: Aleuroplast (protein) Elaioplast (fat) Amyloplast (carbohydrates)

EXAM TRAP!

Prokaryotes → no proper nucleus → called **Nucleoid** (not membrane-bound)

Eukaryotes → true nucleus (membrane-bound) | Mitochondria has own DNA + ribosomes!



② Sports — Relay Race

Q2. A designated portion of the relay race is called a ___?

- ☒ A A leg ✓
- ☐ B A heat
- ☐ C A spike
- ☐ D A lifting leg

✓ Answer: A — Each portion run by one athlete in a relay = "A leg"



③ Ancient History — Vedic Age & Rigveda

Q3. Each Mandala of the Rigveda is further subdivided into?

- ☒ A Suktas (collection of hymns) ✓
- ☐ B Kandaz
- ☐ C Purana
- ☐ D Samhita

✓ Answer: A — Rigveda → 10 Mandalas → each Mandala divided into Suktas (hymns)

VEDIC AGE TIMELINE

Early Vedic / Rigvedic Age: 1500–1000 BC

Later Vedic Age: 1000–500 BC

(Samaveda, Yajurveda, Atharvaveda)

FOUR VEDAS

Rigveda → Oldest; hymns (Suktas); 10 Mandalas; 1028 hymns; recited by Hotri

Samaveda → Music | **Yajurveda** → Rituals (Shukla + Krishna)

Atharvaveda → Magic spells, ward off evil

→ Important Mandalas of Rigveda

Mandala	Importance
3rd	Gayatri Mantra
7th	Battle of Ten Kings (Dasarajna) ; river Ravi (Purushani); most mentions of Saraswati river
9th	Lord Soma (God of Plants); Somarasa (energy drink); Purusha Sukta — how varnas emerged from Brahma
10th	History / Purusha Sukta details

🧠 MEMORY TRICK

Varnas from Purusha (Cosmic Man):

Brahmin → **Mouth** | Kshatriya → **Arms** | Vaishya → **Thighs** | Shudra → **Feet**

Most mentioned rivers: 1. **Saraswati** (later = Ghaggar) | 2. **Sindhu (Indus)**

→ Vedic Literature Structure

UPANISHADS (KEY ONES)

Brihadaranyaka → Largest; in Yajurveda

Mundaka → "Satyameva Jayate" taken from here

Kathopanishad → Dialogue: Nachiketa & Yama

ORTHODOX SCHOOLS (ASTIKA)

Vaisheshika → **Kanada**

Nyaya → **Gautama**

Yoga → **Patanjali**

Sankhya → **Kapil**

Upanishads = at end of every Veda =

Vedanta

Vedanta → **Ved Vyas**

Mimamsa → **Jaimini**

♦ ♦ ♦

④ Polity — Directive Principles of State Policy (DPSP)

Q4. Which is NOT an aim of the DPSP?

- ☐ A Social Democracy
- ☐ B Economic Democracy
- ☒ C Political Democracy ✗ (NOT an aim of DPSP)
- ☐ D Welfare State

✓ Answer: C — Political Democracy is established by Fundamental Rights (Part III), NOT DPSP

DPSP BASICS

Part: **IV** | Articles: **36–51**

Borrowed from: **Ireland**

Inspired by: **Karachi Resolution**

Goal: **Social & Economic Democracy**

Non-justiciable (Article 37) — cannot enforce in court

DPSP CLASSIFICATION

(Not in Constitution — given by scholars)

Socialist | Gandhian |

Liberal/Intellectual

→ Article-wise DPSP — Key Articles

Article	Content
37	Non-justiciability of DPSPs
38(1)	Welfare State
38(2)	Minimize inequality in income/status (Added by 44th Amendment 1978)
39A	Free legal aid to poor (42nd Amendment)
43	Living Wage
43A	Participation of workers in management of industries (42nd Amendment)
44	Uniform Civil Code
45	Early childhood care (Modified by 86th Amendment 2002)

48A	Environment protection (42nd Amendment)
50	Separation of Judiciary from Executive
51	International peace and security

KEY FACTS

- 42nd (1976) → Added 39A, 43A, 48A | 44th (1978) → Added Art. 38 Clause 2
- 86th (2002) → Changed Article 45 | 97th (2011) → Added Art. 43B (Cooperative Societies)



⑤ Biology — Muscle Soreness (Anaerobic Respiration) 🏋️

Q5. Why do muscles feel sore and cramp after intense exercise?

- ☒ A Lactic acid fermentation (Anaerobic respiration) ✓
- ☐ B Aerobic respiration
- ☐ C Glycolysis only
- ☐ D Excess oxygen supply

✓ Answer: A — Intense exercise → insufficient O₂ → lactic acid builds up → soreness/cramps



⑥ Modern History — Subsidiary Alliance (Wellesley) ✂️

Q6. Which statement about Subsidiary Alliance is INCORRECT?

- ☐ A Rulers were free to employ any European ✗ INCORRECT STATEMENT
- ☒ B British promise not to interfere in internal affairs ✓ (this is correct)
- ☒ C Indian ruler had to disband own army ✓ (this is correct)
- ☒ D Had to keep British army and pay for it ✓ (this is correct)

✓ Answer: A — INCORRECT because: Rulers could NOT employ French, Portuguese or other Europeans!

Introduced by **Wellesley (1798–1805)**

Indian ruler MUST:

→ Disband own army

→ Maintain British army at **own expense**

→ Keep permanent **British Resident** in court

→ **NO treaty with any European/Indian** without British permission

→ **CANNOT employ French, Portuguese or Europeans**

1798 → **Nizam of Hyderabad** (FIRST)

1799 → Tanjore + Mysore

1801 → Awadh

After 4th Anglo-Mysore War → Hyder

Ali/Tipu finished → **Wadia Dynasty**

restored

Awadh = **"Nursery of British Army"**

Awadh seized by British in 1856

(Dalhousie — misgovernance)



⑦ Modern History — Indian National Congress

Q7. Who criticized the moderates for their politics of prayer?

(A) Bipin Chandra Pal

(B) Bal Gangadhar Tilak

(C) Lala Lajpat Rai

(D) All of the above (Lal Bal Pal) ✓

✓ Answer: D — All three = "Lal Bal Pal" (the Extremists trio)

→ INC Formation

INC BASICS

Formed: **1885** | Founder: **A.O. Hume**

Viceroy: Lord Dufferin (called Congress "factory of sedition")

1st session: **Gokuldas Tejpal Sanskrit College, Bombay**

1st President: **Womesh Chandra Banerjee** (72 members)

Immediate reason: **Ilbert Bill Controversy (1883)** — Indian judges could try European subjects; British opposed → withdrawn; Viceroy = **Lord Ripon**

LORD RIPON'S KEY ACTS

Repealed **Vernacular Press Act** (1882)

Repealed **Arms Act** (1882)

First **Factory Act** (1881)

Hunter Education Commission (1882)

First synchronized Census (1881)

Known as: **Father of Local Self-Government**

→ INC Sessions – Notable Firsts

Session	Year	President	Notable
1st — Bombay	1885	Womesh Chandra Banerjee	First session
2nd	1886	Dadabhai Naoroji	—
3rd — Madras	1887	Badruddin Tyabji	1st Muslim President
4th — Prayagraj	1888	George Yule	1st Englishman to preside

→ Moderates vs Extremists

MODERATES (PPP Model)

Prayer + Petition + Protest

Dadabhai Naoroji, Gopal Krishna Gokhale,
Womesh Chandra Banerjee, Feroze Shah
Mehta, Surendranath Banerjee

EXTREMISTS (LAL BAL PAL)

Lala Lajpat Rai (Safety Valve theory)

Bal Gangadhar Tilak

Bipin Chandra Pal

Surat Split → 1907 (presided by Ras
Bihari Ghosh)

Reunion → 1916 (Lucknow) — Lucknow
Pact (Congress + Muslim League)

→ Tilak, Besant & Naoroji – Key Facts

Person	Key Facts
Bal Gangadhar Tilak	Newspapers: Maratha (English) + Kesari (Marathi) "Swaraj is my birthright" Called "Father of Indian Unrest" by Valentine Chirol Ganpati + Shivaji festivals (1890s) Home Rule League 1916 (HQ = Pune)
Annie Besant	Home Rule League 1916 (HQ = Madras) First woman president of Congress → 1917
Sarojini Naidu	First INDIAN woman president of Congress → Kanpur session 1925
Dadabhai Naoroji	Presided 3 Congress sessions "Grand Old Man of India" East India Association (1866)



⑧ Medieval History — Alauddin Khilji & Balban



Q8. Primary objective of Alauddin Khilji's market control?

- (A) Replace religious taxation
- (B) Expand treasury by increasing overseas trade
- (C) Keep commodity prices low so soldiers could be maintained on fixed salaries ✓
- (D) Promote overseas markets

✓ Answer: C — Maintained large army; kept price controls so he could afford them at fixed salaries

→ Alauddin Khilji (1296-1316) — Market Reforms

Market	Purpose	Officer
Mandi (Grain Market)	Grain/food	Shahna-e-Mandi
Sarai Adil	Costly goods, clothing	—
Slave Market	Slaves and cattle	—

ALAUDDIN'S OTHER REFORMS

Daag → Branding of horses

Chehra → Hulia system (roll/description of soldiers)

Taxes: **Ghairi** (house) + **Charai** (grazing) + **Kharaj** (land = 50%)

Abolished **Iqta system** (started by Iltutmish)

Built: **Alai Darwaza** (entrance to Qutub Minar)

Khilji dynasty founder: **Jalaluddin Khilji (1290-96)**

Alauddin murdered Jalaluddin in Allahabad jail

AMIR KHUSRAU (Court of Alauddin)

Known as: **Tuti-e-Hind (Parrot of India)**

Works: Tughlaqnama, Nuh Sipih, Khazain-ul-Futuh

Invented: **Sitar and Tabla**

Malik Kafur = Military commander = **Hazar Dinari**

→ Balban (1266-1287)

- Started: **Prostration (Sijda) + Paibos (kissing feet)** | Policy: **Blood and Iron**
- Killed **Tughril Beg** (governor of Bengal who rebelled)
- Abolished **Chahalgani/Chalisa** (Group of 40 slaves) started by Iltutmish
- Declared himself descendant of **Afrasiyab**
- Established **Diwan-i-Arz** (Military department)
- Started **Navroz festival** (Persian New Year; Spring Equinox)

⑨ Geography — Plate Tectonics & Oceans

Q1. Mid-ocean ridges are formed at ___ plate boundaries?

- ☒ A Divergent ✓
- ☐ B Convergent
- ☐ C Transform
- ☐ D Passive

✓ Answer: A — Divergent = plates move apart → magma fills gap → mid-ocean ridges

Boundary	Movement	Result	Nature
Convergent	Plates collide; denser subducts	Subduction zone; volcanoes; mountains	Destructive
Divergent	Plates move apart; magma fills gap	Mid-ocean ridges; sea floor spreading	Constructive
Transform	Plates slide past each other	Earthquakes; faults	Conservative

🔑 KEY FACTS

- 🔑 Sea Floor Spreading → **Harry Hess (1961)** | Convection Currents → **Arthur Holmes**
- 🔑 Total major plates = **7** | Largest plate = **Pacific Plate**
- 🔑 Mid-Atlantic Ridge = Divergent | Pacific Ring of Fire = **Convergent** | San Andreas Fault = Transform

→ Atlantic Ocean & World Straits

ATLANTIC OCEAN FACTS

Shape: **S-shaped** | Busiest ocean
Sargasso Sea → North Atlantic; no land border; **Sargassum** (brown algae)
 Other brown algae: Kelp, Laminaria
Newfoundland, Canada → Labrador Current (cold) + Gulf Stream (warm) meet → Best fishing + Foggy

WORLD STRAITS

Panama Canal → Pacific ↔ Atlantic
Bering Strait → Arctic ↔ Pacific; Asia ↔ America
Strait of Hormuz → Persian Gulf ↔ Gulf of Oman
Bab-el-Mandeb → Red Sea ↔ Gulf of Aden = "Gateway of Tears"
Strait of Gibraltar → Atlantic ↔ Mediterranean

Strait of Malacca → Andaman Sea ↔
South China Sea

Grassland	Location
Campos	Brazil
Llanos	Venezuela
Pampas	Argentina

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⑩ Economics — Government Deficits

Q10. Which indicator best captures borrowing requirement **EXCLUDING** interest payments?

- ☐ A Budget Deficit
- ☐ B Fiscal Deficit
- ☐ C Revenue Deficit
- ☒ D **Primary Deficit** ✓

✓ Answer: D — $\text{Primary Deficit} = \text{Fiscal Deficit} - \text{Interest Payments}$

FORMULA

Fiscal Deficit = Total Expenditure – (Revenue Receipts + Non-debt creating Capital Receipts)

Primary Deficit = Fiscal Deficit – Interest Payments (borrowing **EXCLUDING** past debt interest)

Revenue Deficit = Revenue Expenditure – Revenue Receipts

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⑪ History — Temple Architecture

Q11. What historical process is best reflected by evolution of temple architecture?

- ☒ A Cultural interaction and synthesis across regions and dynasties ✓
- ☐ B Decline of cultural interaction ✗
- ☐ C Only royal patronage
- ☐ D Religious uniformity

✓ Answer: A — Temple architecture EVOLVED through cultural interaction, NOT declined

Style	Region	Developed By
Nagara	North India	Guptas
Dravidian	South India	Pallavas
Vesara	Central India	Chalukyas

Feature	Nagara (North)	Dravidian (South)
Main tower	Shikhara	Vimana
Platform	Jagati (square)	High platform
Entrance gateway	None (no gopuram)	Gopuram (tall gateway)
Inner sanctuary	Garbhagriha	Garbhagriha
Pillared hall	Mandapa	Mandapa

→ Regional Variations

KALINGA ARCHITECTURE (ODISHA)

Main temple tower: **Rekha Deula**
Mandapa: **Jagamohana** (= Peedha Deula)
Konark Sun Temple → built by
Narasimha Deva I
Known as: "**Black Pagoda**"
Structure: Chariot with **7 horses + 24 wheels**
(24 wheels = 24 hours / 8 spokes + 8 sub-spokes)

HOYSALA (KARNATAKA)

Plan: **Stellate (star-shaped)**
Capital: **Halebidu (Dwarasamudra)**
Famous temples: Hoysaleswara,
Chennakeshava

⑫ Physics — Mass vs Weight

Q12. Weight of 45 kg mass ($g = 9.8$)?

- ☒ A 441 N (≈ 44.1 kgf) ✓
- ☐ B 45 N
- ☐ C 450 N
- ☐ D 4.5 N

✓ Answer: A — $\text{Weight} = \text{Mass} \times g = 45 \times 9.8 = 441$ N

Property	Mass	Weight
Type	Scalar	Vector
Unit	kg	N (Newton)
Depends on gravity?	No	Yes
On Moon	Same as Earth	1/6th of Earth

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⑬ Polity — Parliament & Sessions

Q13. Which Article deals with sessions of Parliament?

- ☒ A Article 85 ✓
- ☐ B Article 83
- ☐ C Article 79
- ☐ D Article 100

✓ Answer: A — Article 85 = Sessions of Parliament

LS vs RS COMPARISON

Feature	Lok Sabha	Rajya Sabha
Min. age	25 years	30 years
Duration	5 years	Permanent
Member term	5 years	6 years
Retirement	—	1/3 every 2 yrs

SESSIONS OF PARLIAMENT

Three sessions: **Budget** (longest) + **Monsoon** + **Winter**

Gap between sessions = **Recess** (max 6 months)

Minimum **2 sessions per year** mandatory

Summoning + Prorogation → **President**

Adjournment → **Presiding Officer**

Dissolution → President dissolves **ONLY**
Lok Sabha

→ Important Articles – Parliament

Article	Content
74	Council of Ministers headed by PM to aid & advise President
78	Duties of PM (link between CoM and President)
79	Composition of Parliament (LS + RS + President)
85	Sessions of Parliament
87	Special Address of President — TWO occasions: (1) First session after General Elections; (2) First session of every financial year
89	VP as ex-officio Chairman of Rajya Sabha
93	Speaker and Deputy Speaker of Lok Sabha

→ Article 100 – Voting & Quorum

KEY FACTS

- Quorum = **1/10th** of total members (Clause 3)
- If removal resolution against **Speaker** → Speaker sits as normal member (not presiding)
→ has **first instance vote only**
- VP (Rajya Sabha Chairman) → **NO first instance vote** (not a member of any house)
→ has only casting vote
- If removal resolution against VP → **NO voting at all** (neither first instance nor casting)
- PM = **de facto / real head** | President = **de jure / nominal / titular head**

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⑭ Polity — All 12 Constitutional Schedules

Schedule	Content
1st	States and Union Territories
2nd	Emoluments for President, Governors, Judges, CAG, Speaker etc. (NOT MPs/MLAs)
3rd	Oaths and Affirmations (MPs, MLAs, Ministers, Judges)

4th	Allocation of seats in Rajya Sabha
5th	Scheduled Areas and Scheduled Tribes
6th	Tribal Areas — ATM-M: Assam, Tripura, Meghalaya, Mizoram
7th	Union List + State List + Concurrent List
8th	Languages — originally 14; now 22 (Sindhi by 21st Amend; Konkani/Nepali/Manipuri by 71st; Bodo/Dogri/Santhali/Maithili by 92nd)
9th	Abolition of Zamindari laws (First Amendment, Nehru's time); IR Coelho Case — laws before April 24, 1973 exempt from judicial review
10th	Anti-Defection Law (52nd Amendment, 1985); Raghav Chadha case
11th	Panchayat — 29 functional matters (73rd Amendment)
12th	Municipalities — 18 functional matters (74th Amendment)

EXAM TRAP!

2nd Schedule = emoluments for President/Judges etc. **NOT for MPs/MLAs** (they're covered separately)!

Judicial Review borrowed from → **USA**



⑮ Geography — Himalayan Passes

Q15. Statements: (1) Zojila connects Kashmir Valley and Ladakh ✓ | (2) Shipkila Pass is in Arunachal Pradesh ✗

C Only Statement 1 is correct ✓

✓ Answer: C — Shipkila is in HIMACHAL PRADESH, NOT Arunachal Pradesh!

EXAM TRAP!

Shipkila Pass → Himachal Pradesh (where Sutlej river enters India) — NOT Arunachal Pradesh!

Pass	State	Key Fact
Pir Panjal / Banihal	J&K	Jammu ↔ Srinagar

Zojila (Zaskar Range)	J&K	Srinagar ↔ Leh/Ladakh
Burjil	Ladakh Range	Srinagar → Gilgit-Baltistan (PoK)
Rohtang	Himachal Pradesh	Manali ↔ Lahaul-Spiti Ravi & Beas rivers originate
Bara-lacha-la	Himachal Pradesh	Lahaul-Spiti ↔ Leh Chenab river originates (Chandra + Bhaga meet at Tandi village)
Shipkila	Himachal Pradesh	Sutlej river enters India here (originates near Rakas Lake, Tibet)
Bhumla, Bomdila, Diphu	Arunachal Pradesh	Diphu = tri-junction India + Tibet/China + Myanmar

RIVER FACTS

Chenab = largest tributary of Indus | **Sutlej** = longest tributary of Indus (Suturi; in Tibet = Longchen Khamb)

Himalayan range order (W→E, N→S): Trans-Himalayas: Karakoram → Ladakh → Zaskar → Great Himalayas → Lesser Himalayas → Shivalik

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16 Economics — Human Development Index (HDI)

Q16. Which indicator measures the KNOWLEDGE dimension of HDI?

- ☒ A Mean years of schooling + Expected years of schooling ✓
- ☐ B Life expectancy at birth
- ☐ C GNI per capita
- ☐ D Literacy rate

✓ Answer: A — Knowledge dimension = Mean years of schooling + Expected years of schooling

HDI BASICS

Published by: **UNDP**

Concept introduced: 1990 by **Mahbub ul**

Haq (Pakistani economist)

Role of **Amartya Sen** (Nobel Economics 1998 — first Indian)

THREE HDI DIMENSIONS

Long & Healthy Life → **Life Expectancy at Birth**

Knowledge → **Mean Years + Expected Years of Schooling**

Decent Living → **GNI per capita (PPP)**

→ MPI & MVI

Index	Dimensions	Indicators
Global MPI	Standard of Living + Health + Education	10 indicators (UNDP/OPHI)
National MPI (India)	Same 3 dimensions	12 indicators (+ Bank account + Prenatal care)
MVI	Vulnerability + Resilience	Each has 3 components: Social + Economic + Environmental

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⑰ Economics — CRR & Money Multiplier

Q17. Higher CRR ___ the Money Multiplier?

- ☐ A Increases
- ☒ B Reduces ✓
- ☐ C No effect
- ☐ D Doubles

✓ Answer: B — Higher CRR → banks lend less → less money circulates → Money Multiplier decreases

Ratio	Full Form	Kept Where	Interest?
CRR	Cash Reserve Ratio	Cash with RBI	No interest earned
SLR	Statutory Liquidity Ratio	Govt securities/gold	Interest earned

NDTL = NET DEMAND AND TIME LIABILITIES

Demand liabilities = Savings + Current accounts

Time liabilities = Fixed deposits

Spread = Banks' profit = loan interest earned – deposit interest paid

Higher CRR → Contractionary effect (less money in economy)



18 Economics — Government Schemes (Agriculture)

Q18. Which pairs (scheme-ministry) are correctly matched?

- 1 PM-AASHA — Consumer Affairs ✗ (correct: Ministry of Agriculture)
- 2 eNAM — Ministry of Agriculture & Farmers Welfare ✓
- 3 PMKSY — Ministry of Jal Shakti ✓
- 4 RoDTEP — Ministry of Finance ✗ (correct: Commerce & Industry)

✓ Answer: B — Only 2 and 3 are correctly matched

Scheme	Correct Ministry
PM-AASHA	Ministry of Agriculture & Farmers Welfare (NOT Consumer Affairs)
eNAM (Electronic National Agriculture Market)	Ministry of Agriculture & Farmers Welfare ✓
PMKSY (PM Krishi Sinchai Yojana)	Ministry of Jal Shakti ✓
RoDTEP	Ministry of Commerce & Industry (NOT Finance)



19 Geography — Seasons, Solstice & Equinox

Q19. Navroz (Parsi festival; started by Balban) is celebrated on which event?

- A Spring (Vernal) Equinox — March 21st ✓
- B Summer Solstice — June 21st
- C Winter Solstice — December 22nd
- D Autumn Equinox — September 23rd

✓ Answer: A — Navroz = Spring/Vernal Equinox (March 21) = Persian New Year

Date	Position	Name	Significance
June 21	Sun overhead Tropic of Cancer (23.5°N)	Summer Solstice	Longest day in Northern Hemisphere
December	Sun overhead Tropic of Capricorn	Winter Solstice	Shortest day in Northern Hemisphere

22	(23.5°S)		Hemisphere
March 21	Sun overhead Equator	Spring/Vernal Equinox	Equal day & night everywhere
September 23	Sun overhead Equator	Autumn Equinox	Equal day & night everywhere

EARTH-SUN DISTANCE

Perihelion (closest) = 147 million km = **January 3-4**

Aphelion (farthest) = 151 million km = **July 4**

(Closest to Sun in winter → counter-intuitive!)

LATITUDE LINES

66.5°N → **Arctic Circle**

23.5°N → **Tropic of Cancer**

0° → **Equator**

23.5°S → **Tropic of Capricorn**

66.5°S → **Antarctic Circle**

Frigid Zones → 6 months day/6 months night

♦ ♦ ♦

20 Economics — Liquidity 💰

Q20. Which has highest liquidity?

- ☒ A Cash ✓
- ☐ B Gold
- ☐ C Fixed Deposits
- ☐ D Real estate

✓ Answer: A — Cash = most liquid asset | Gold = valuable but NOT most liquid

EXAM TRAP!

Liquidity Order: Cash > Savings account > Fixed Deposits > Gold > Real estate

Gold is valuable but requires time to convert to cash → less liquid than cash!

♦ ♦ ♦



Teacher's Study Tips — Memorise These!



STUDY TIP



STUDY TIP

1. **Active Recall** → Don't just read; pause and try to remember everything about a topic
2. **Note Pain Points** → If stuck at Article 47 or 48, note it and revise from Fat Man notes
3. After GK session (11:30) → **Rest 15-20 min, NO phone** → let knowledge consolidate
4. **Brain rest** → Eyes closed / lying down; drink water → helps retention

5. **First round in exam** → Attempt easy questions first (builds confidence); tough ones later
6. **Don't memorise formulas** → Understand them! (e.g., fiscal deficit = borrowing = expenditure minus non-borrowed receipts)
7. **GK time (10-11:30)** → Dedicate entirely to GK; full immersion; no other thoughts



Quick Revision Flash Table — All 20 Questions

Q.No	Question	Answer
Q1	Golgi apparatus function?	Protein modification, packaging, dispatching = "Post Office of Cell"
Q2	Portion of relay race?	A leg
Q3	Mandala of Rigveda divided into?	Suktas 10 Mandalas 1028 hymns
Q4	NOT aim of DPSP?	Political Democracy (DPSP → Social + Economic democracy)
Q5	Muscle soreness cause?	Lactic acid fermentation (Anaerobic respiration)
Q6	Incorrect about Subsidiary Alliance?	Rulers CANNOT employ Europeans (French/Portuguese etc.)
Q7	Who criticized moderates' prayer politics?	Lal Bal Pal — All three (Lala Lajpat Rai + Tilak + Bipin Chandra Pal)
Q8	Alauddin Khilji's market control objective?	Keep prices low to maintain soldiers on fixed salaries
Q9	Mid-ocean ridges formed at?	Divergent plate boundaries (Constructive)
Q10	Borrowing req. excluding interest?	Primary Deficit = Fiscal Deficit – Interest Payments
Q11	Evolution of temple	Cultural interaction and synthesis (NOT decline)

	architecture reflects?	
Q12	Weight of 45 kg ($g = 9.8$)?	441 N Mass = Scalar Weight = Vector
Q13	Article for sessions of Parliament?	Article 85 Summoning + Prorogation = President Adjournment = Presiding Officer
Q14	Schedule 10 content?	Anti-Defection Law (52nd Amendment 1985) 12th Schedule = Municipalities
Q15	Shipkila Pass is in which state?	Himachal Pradesh (NOT Arunachal) Sutlej enters India here
Q16	Knowledge dimension of HDI?	Mean years + Expected years of schooling HDI by UNDP (Mahbub ul Haq 1990)
Q17	Higher CRR → Money Multiplier?	Reduces CRR = cash with RBI, no interest SLR = govt securities, interest earned
Q18	Correct scheme-ministry pairs?	eNAM + PMKSY only PM-AASHA = Agriculture (NOT Consumer Affairs)
Q19	Navroz celebrated on?	Spring/Vernal Equinox (March 21) Started by Balban
Q20	Highest liquidity asset?	Cash (NOT gold — gold valuable but less liquid)

 END OF MOCK TEST #4 NOTES

All 20 Questions Covered  | Apply the study tips. You've got this! 